

Chapter 1: Networking Devices

Network Devices and Configurations

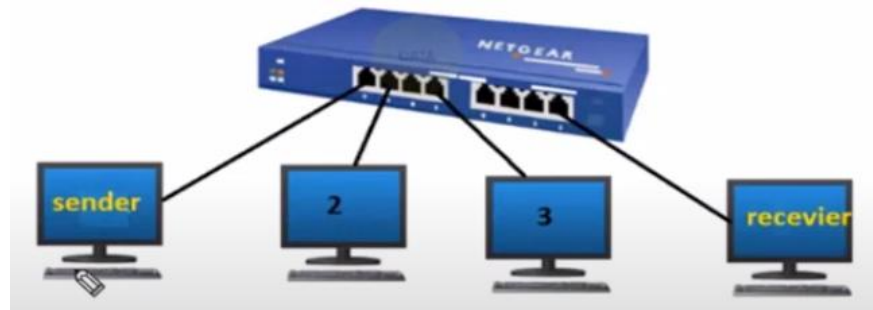
What to cover

- Hub
- Repeater
- Bridge
- Switch
- Layer 2 Switch
- Router
- Gateway
- Firewall
- Access Points
- Wireless LAN Controllers
- Network Interface Card (NIC)

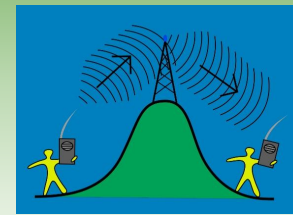


Hub

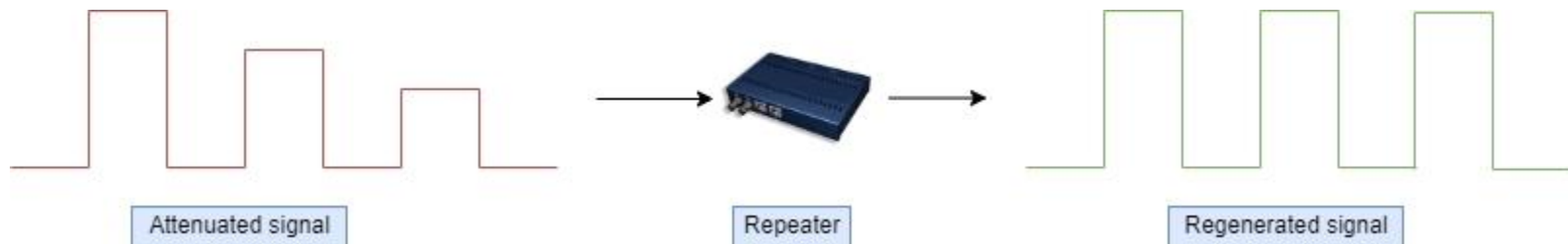
- A networking device that operates at the **physical** layer.
- Connects devices together to share data.
- It is old technology not commonly used.
- **Disadvantage:** consumes a lot of bandwidth due to broadcasting (not intelligent).
- **How It Works** - When hub receives data from one device, the data is sent to all (broadcast) other devices connected to it.



Repeater



- Repeaters are network devices operating at **physical layer** of the **OSI model** that amplify or regenerate an incoming signal before retransmitting it.
- They are incorporated in networks to expand its coverage area.
- They are also known as signal boosters.
- Connects similar networks.



Bridges

- Bridges are used to connect two subnetworks that use interchangeable protocols.
- It combines two LANs to form an extended LAN.
- The main difference between the bridge and repeater is that the bridge has a penetrating efficiency.

u *How IT Works*

- A bridge accepts all the packets and amplifies all of them to the other side. The bridges are intelligent devices that allow the passing of only selective packets from them. A bridge only passes those packets addressed from a node in one network to another node in the other network.

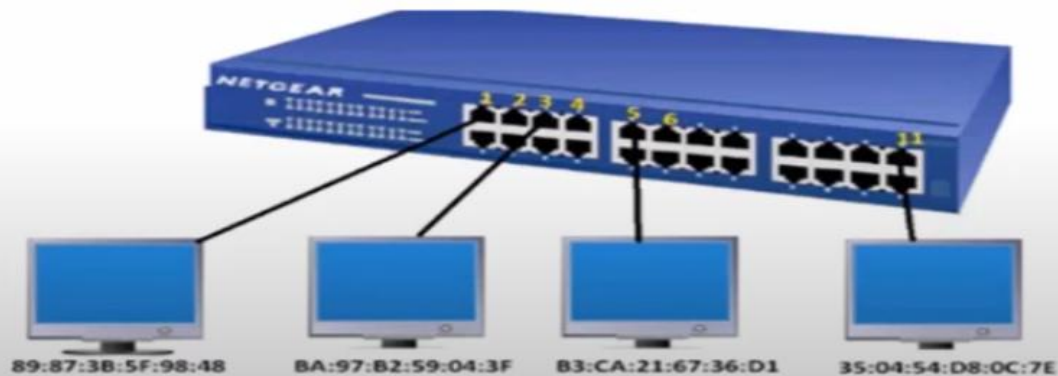


Switch

- A networking device that operates at the **data link** layer of OSI model.
- Connects devices together to share data.
- Uses MAC- address to send and receive data.
- Sends data for only specific device, new technology, conserves bandwidth.

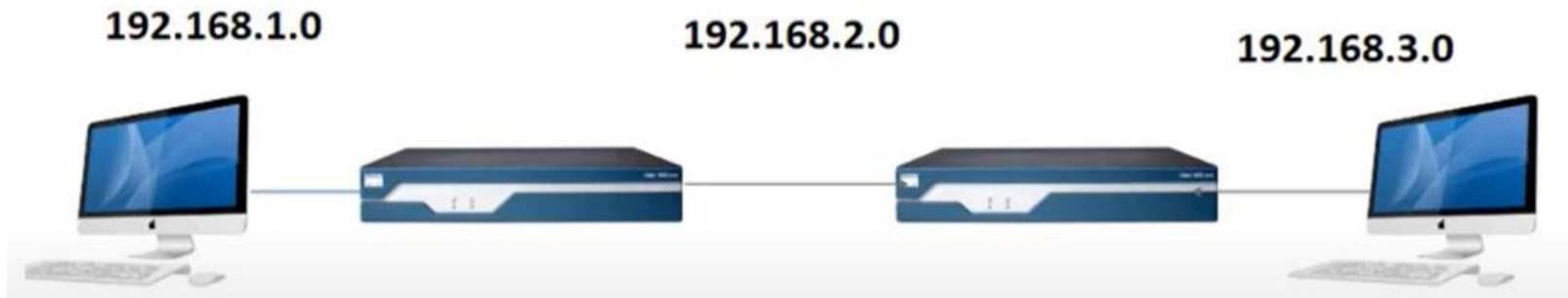
How It Works – Switch has a memory called Content Addressable Memory (CAM) table or MAC-Address table. The table stores the device MAC- address corresponding to the switchport numbers it is connected. When data is sent from one device to another, upon reaching the CAM table the switch checks destination MAC-address of the data and forwards the data to the port number that corresponds the destination MAC-address.

Switch port numbers	Computers MAC address
1	89:87:3B:5F:98:48
3	BA:97:B2:59:04:3F
5	B3:CA:21:67:36:D1
11	35:04:54:D8:0C:7E



Router

- A network device that operates at the network layer of OSI model.
- Connects multiple or different networks.
- Performs packet directing functions in the network.
- Uses IP addresses to send and receive data.
- Assigns IP addresses to devices.
- Chooses best path for routing.



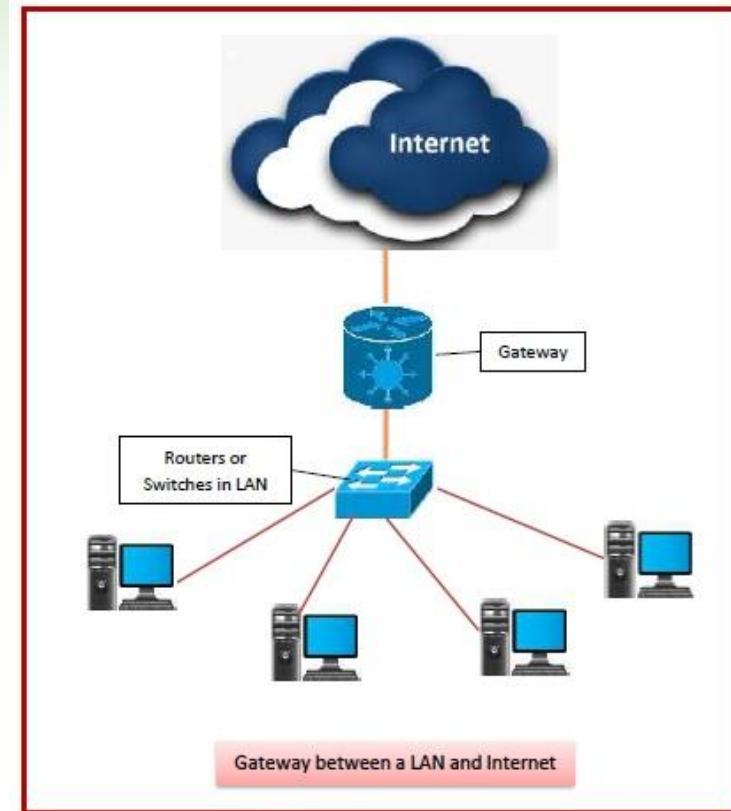
Layer 3 Switch

- A network device that operates both at the data link layer and network layer of OSI model.
- Works both as a switch and router.
- Supports both switching and routing technologies.
- Works with same and different networks.
- Also called multilayered switch.



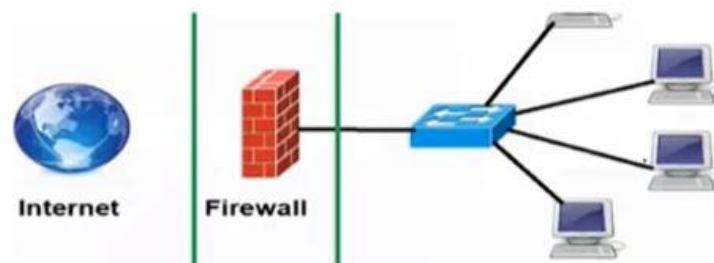
Gateways

- A gateway is a network node that forms a passage between two networks operating with different transmission protocols.
- The most common type of gateways, the network gateway operates at layer 3, i.e. **network layer** of the OSI model.
- However, depending upon the functionality, a gateway can operate at any of the seven layers of **OSI model**.
- It acts as the **entry – exit** point for a network since all traffic that flows across the networks should pass through the gateway.
- Only the internal traffic between the nodes of a **LAN** does not pass through the gateway.



Firewall

- Can be a hardware or software that provides security to the network..
- Filters (Allow or deny) the traffic coming in or going out of the network.
- Protect against malware.
- Stops hackers to some extent.



Types of Firewall

Packet filtering – also called Access Control List;

- If packet match, it's allowed. If packet don't match. It is blocked.
- Checks source and destination IP plus source and destination ports.
- Cannot check payload (data inside the packet).

Proxy – also called application firewall;

- It's a software that you download and install e/e. FortiGate, Norton, Avast Endpoint protection, etc.
- Checks source and destination IP plus source and destination ports.
- Checks payload (data inside the packet).

Hybrid – Combines the functionalities of both packet filtering and proxy firewalls.

- The best of all.



Access Points (AP, WAP)

- This is a network device that creates WIFI or wireless network.
- Also known as wireless access point.
- They enable users to connect through the internet.
- The signal is stronger when you are near the device but weaker as you move further



Wireless LAN Controller (WLC)

- Also called Wireless Controller
- This is a networking device that controls many access points in a network.
- All the access points are connected to the WLC.
- Mostly used where there are many access points.



Network Interface Card (NIC)

- A network interface card (NIC) is a hardware component without which a computer cannot be connected over a network.
- Provides a dedicated network connection to the computer.
- Also called Network Interface Controller, network adapter, or LAN adapter.
- NIC is both a physical layer and a data link layer device.,
- NIC allows both wired and wireless communications.
- NIC allows communications between computers connected via local area network (LAN) as well as communications over large-scale network through Internet Protocol (IP).



Internal/Wired



Wireless



DSU/CSU (Data Service Unit/Channel Service Unit)

- A Digital Service Unit (DSU) is a piece of hardware that connects a Local Area Network (LAN) to an external communication carrier service via a Channel Service Unit (CSU).
- A DSU is a device with two or more ports, one designated as the WAN port and the other as the DTE port.
- The device translates bipolar digital signals from a digital circuit and a CSU that is compatible with the data terminal equipment to which the data is transferred.
- When information is sent from the DTE to the circuit, the DSU performs a similar operation in reverse.
- The telecommunications service that a DSU supports can be a point-to-point or multipoint operation in a digital environment.



Networking Devices and Configuration

End of *Chapter one* !!!

Thank you, Have a nice day!!

Never Stop Learning!

